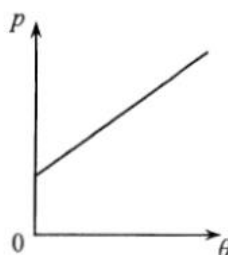
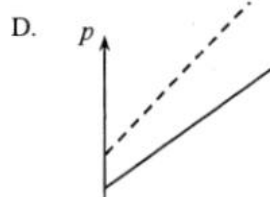
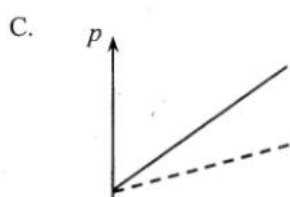
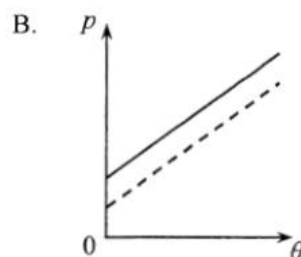
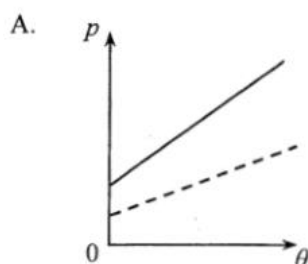


2012 Q3

- *3. An ideal gas is contained in a closed vessel of fixed volume. The graph below shows the variation of pressure p of the gas against its Celsius temperature θ .

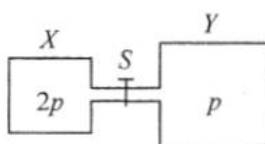


If the number of gas molecules in the vessel is halved, which graph of the dotted line best shows the relationship between p and θ ?



2013 Q4

- *4.



Vessel X of volume V and vessel Y of volume $2V$ are connected by a short narrow tube as shown. Initially, tap S is closed and the same kind of ideal gas at the same temperature is contained in X and Y at pressure $2p$ and p respectively. The tap S is then opened and equilibrium state is finally reached with the temperature unchanged. Which statement is **INCORRECT**?

- A. Before S is opened, both vessels contain the same number of gas molecules.
- B. Before S is opened, the average kinetic energy of the gas molecules in both vessels is the same.
- C. When S is opened, a net flow of gas from X to Y occurs.
- D. When equilibrium is reached, the gas pressure becomes $\frac{3}{2}p$.

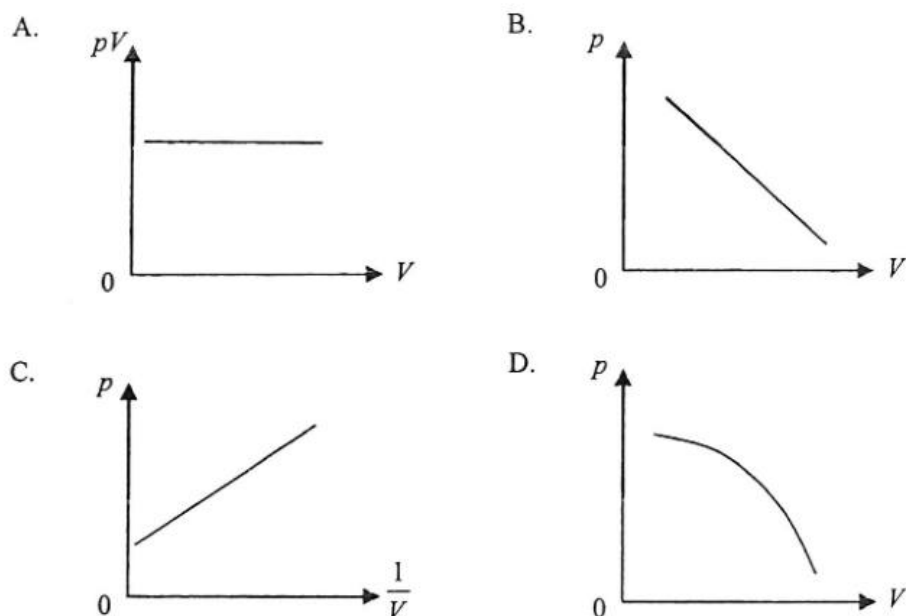
2017 Q4

- *4. The pressure of a fixed mass of an ideal gas at 10°C is $2 \times 10^5 \text{ N m}^{-2}$. If the volume of the gas is reduced to half of its original volume and its temperature is increased to 100°C , what would the pressure be ?

- A. $1.00 \times 10^5 \text{ N m}^{-2}$
- B. $1.32 \times 10^5 \text{ N m}^{-2}$
- C. $4.00 \times 10^5 \text{ N m}^{-2}$
- D. $5.27 \times 10^5 \text{ N m}^{-2}$

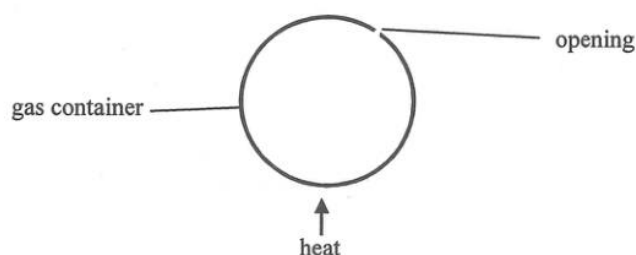
2018 Q4

- *4. From which graph below can one deduce that the pressure p of a fixed mass of an ideal gas is inverse proportional to its volume V when the temperature of the gas is kept constant ?



2019 Q3

- *3. The figure shows an inexpandable container with an opening.



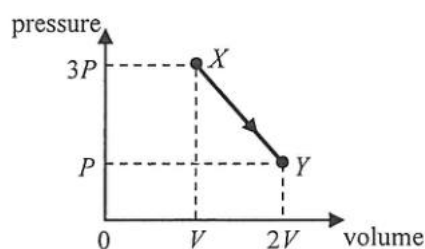
When the gas inside the container is being heated slowly by a heater, which statements about the gas molecules in the container are correct ?

- (1) The number of molecules decreases.
- (2) The average kinetic energy of molecules increases.
- (3) The average separation between molecules remains unchanged.

- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

2020 Q3

- *3. A fixed mass of an ideal gas expands from state X to state Y through a process as represented in the pressure-volume graph below.



If the temperature of the gas at state Y is 25°C , what is its temperature at state X ?

- A. -74.3°C
- B. 16.7°C
- C. 37.5°C
- D. 174°C

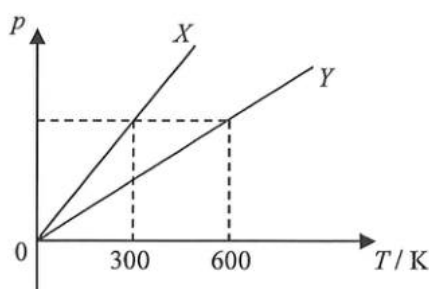
2022 Q3

- *3. A weather balloon filled with 21 kg of helium gas has a volume of 120 m^3 at 27°C . Find the gas pressure in the balloon. Given: mass of one mole of helium gas = $4 \times 10^{-3}\text{ kg}$

- A. $9.93 \times 10^4\text{ Pa}$
- B. $1.00 \times 10^5\text{ Pa}$
- C. $1.05 \times 10^5\text{ Pa}$
- D. $1.09 \times 10^5\text{ Pa}$

2023 Q3

- *3. A fixed mass of an ideal gas contained in a closed vessel X of volume V is heated. Line X in the figure shows the variation of pressure p with temperature T of the gas. The experiment is repeated with another closed vessel Y containing the same amount of this ideal gas and line Y represents the result.



Find the volume of vessel Y .

- A. $0.25 V$
- B. $0.5 V$
- C. $2V$
- D. $4V$